

From Signals to Species: Improving Acoustic Classification Reliability for North Pacific False Killer Whales

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1 This study presents an automated pre-classification data filtering method that increases
2 confidence in the classification of cetacean acoustic events lacking visual verification, while
3 minimizing bias and reducing the need for manual review. We test the approach on false killer
4 whale (*Pseudorca crassidens*) acoustic data from the North Pacific Ocean. A minimum data
5 requirement derived from sensitivity analyses, combined with filters based on species-specific
6 acoustic characteristics, define objective criteria for excluding acoustic events before
7 classification. This streamlined, reproducible framework simplifies the classification problem
8 and can be adapted to other species to support ongoing population assessments and conservation
9 efforts.

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1. Introduction

Monitoring cetacean populations over time and space to understand trends in abundance, distribution, and ecology are critical for effective conservation and management in the face of environmental change and anthropogenic activities. Passive acoustic monitoring (PAM) is an established cetacean research method for detecting species-specific vocalizations, offering additional data to inform management decisions (Ross et al., 2023; Van Parijs et al., 2009). The widespread use of PAM methods has helped fill key knowledge gaps, particularly for species that exhibit cryptic surface behavior, forage at depth for extended periods (McCullough et al., 2021), inhabit remote and logistically challenging regions (Rice et al., 2021), and serve as a valuable complement to traditional visual observation methods (Rankin et al., 2008). However, the utility of PAM data depends on accurately attributing the vocalizations to the correct species. Acoustic detections with visual verification and animal-borne acoustic recording tags (Johnson et al., 2009; Sayigh et al., 2016) have been instrumental in confirming the species identity of cetacean vocalizations to support PAM research efforts, such as species detection and classification (Bittle & Duncan, 2013).

False killer whales (*Pseudorca crassidens*) in Hawaiian waters are challenging to study using traditional visual survey methods because their high mobility, offshore distribution, and their tendency to occur in small, dispersed subgroups limit detectability (Bradford et al., 2014). However, their highly vocal behavior makes them well-suited for PAM, which can improve detection compared to visual methods alone (Gruden et al., 2023). If false killer whales can be confidently identified using PAM data, this would contribute an additional data source to inform management actions for this protected species. Previous acoustic classification models relied exclusively on either false killer whale echolocation clicks or whistles, resulting in mixed

accuracy and requiring significant manual annotation or validation of the test or training data (Baumann-Pickering et al., 2015; Oswald et al., 2007; Ziegenhorn et al., 2022). Recent advances in machine-learning classification models have incorporated automatically detected whistles and clicks into a binary hierarchical random forest model to accurately distinguish false killer whale signals from those of other pooled odontocete species, producing a 99% overall correct classification score (McCullough et al., 2021).

Identifying species within a PAM dataset is essential for downstream analyses, such as abundance estimation and species distribution models (Davis et al., 2017; Marques et al., 2013), and can meaningfully increase observations of data-limited species, such as false killer whales. Manual review methods can be used to verify the presence of some species in PAM datasets based on distinct features of acoustic signal, assessing vocalization patterns, or reviewing localization data when available. While these steps ensure accuracy by reducing false positives and false negatives that would impact subsequent analyses, they are time-consuming for larger datasets and results can be subject to bias and variability introduced by human analysts (Širović, 2016). As PAM datasets continue to grow, so do the automated detection and classification algorithms that can reduce the time and effort required to analyze PAM data compared to manual methods (Kowarski & Moors-Murphy, 2021; Usman et al., 2020). Overall, automated approaches improve objectivity, generating consistent and reproducible results that support further quantitative analyses (Leroy et al., 2018; Miller et al., 2023).

We present a set of pre-classification data filtering steps to enhance the reliability of automated false killer whale acoustic classification results, particularly from large PAM datasets lacking visual verification. First, a minimum data requirement is established based on precision and recall metrics from a sensitivity analysis. Second, species-specific acoustic data filters are

developed using acoustic characteristics measured from visually-verified PAM data. We evaluate these pre-classification steps using an independent dataset as a case study, in which the acoustic events are filtered and then classified using the binary false killer whale model developed in McCullough et al. (2021). Results demonstrate that pre-classification filtering reduces false positives compared to classification performed on unfiltered acoustic events. This improves the reliability of classification results with lower scores, decreases the amount of data requiring manual review, and supports the inclusion of acoustic events without visual verification in downstream analyses. Although developed for false killer whales, this approach is readily adaptable to other cetacean species and acoustic classification models.

2. Methods

2.1 Data collection

The PAM data used to develop and test the pre-classification data filtering steps were collected using towed hydrophone arrays during shipboard visual and acoustic line-transect surveys for cetacean abundance between 2010 and 2024. Towed arrays consisted of two or more hydrophones capable of detecting odontocete signals (echolocation clicks, whistles, and burst pulses) across a 2–85 kHz frequency range. From 2010 - 2012, arrays used EDO EC65 hydrophones (2 - 40 kHz, flat frequency response); in 2013, APC 42-1021 hydrophones were used (2 - 40 kHz, linear decrease in sensitivity of 35 dB); and from 2015 - 2024, HTI-96-MINs were implemented (2 - 100 kHz, flat frequency response). More information about each survey can be found in Table S1 in the supplementary materials.

During real-time data collection, acoustic events were identified if three or more signals were detected, with at least one having an associated bearing angle relative to the towed array.

Some acoustic events were confirmed to species based on concurrent visual observations by trained marine mammal observers (visually-verified events). Events with no visual verification (acoustic-only events) were initially grouped into broad categories based on the general frequency range of signals present (e.g., unidentified small, medium, or large dolphins). Beaked whales (Ziphiidae) and sperm whales (*Physeter macrocephalus*) could be identified to species in real-time based on the species-specific spectral and temporal characteristics of their detected clicks (Backus & Schevill, 1966; Baumann-Pickering et al., 2013; Zimmer et al., 2005).

Initial post-survey processing included extracting acoustic signals from acoustic events following the methods described in McCullough et al. 2021, which included three PAMGuard detectors: the click detector that was divided into six click classification categories based on peak frequency, a whistle and moan detector, and a cepstrum detector for burst pulse signals (Gillespie et al., 2026). Signal features were extracted using R package *PAMpal* (Sakai, 2020). The processed datasets included all visually-verified events of a single species as well as acoustic-only events that only included a single category of unidentified dolphins. All mixed-species events and portions of overlapping events where other odontocetes were visually sighted or acoustically localized within 3 nmi were excluded.

McCullough et al. (2021) used the visually-verified acoustic events collected during the 2017 survey (Table S1) to test and train a binary acoustic classification model. The model development followed the two-step, random forest framework using the R package, *banter* (Rankin et al., 2017) to configure a binary classifier, hereafter referred to as the Pc-vs-Other model, that distinguished false killer whales from 9 other odontocete species combined with high accuracy. Given the high performance of the Pc-vs-Other model, we used it to develop and evaluate one of the pre-classification data filters (Section 2.2.1). We then applied the Pc-vs-Other

model to an independent dataset of acoustic events before and after filtering to compare resulting classification scores and assess filter performance (Section 2.3).

2.2 Pre-classification data filters

2.2.1 Sensitivity analysis to establish a minimum-data filter

The number of signals detected varies between acoustic events, but the amount of data needed to reliably classify acoustic-only events to false killer whales is not well-understood. Therefore, we established a minimum-data filter (MDF) using a sensitivity analysis to remove acoustic events that did not have enough data, and thus information, to be classified with confidence. This analysis evaluated how signal quantity and type affect classification performance by subsampling signals from a set of visually verified acoustic events (described below). First, 15 minimum-data values were selected *a priori* to represent the different numbers of randomly-selected signals to be tested, ranging from 3 (the minimum required in the field to qualify as an acoustic event) to 30,000 signals per detector. These subsets were then classified by the Pc-vs-Other model to predict species identity. This process was repeated 100 times for each minimum-data value to evaluate the performance across the varying subsets of signals.

To establish the MDF, the sensitivity analysis utilized a training dataset of visually verified acoustic events consisting of 33 false killer whale events and 237 events of other pooled odontocete species. These events were identified to species by expert observers during line-transect cetacean surveys in the Central and Western Pacific conducted from 2010-2016 and 2018-2021 (Table S1). The effects of signal type (e.g., clicks vs. whistles) on classification performance were also examined by running the sensitivity analysis on single signal types and various combinations of signal types. Given the class imbalance between false killer whales and

other pooled species, performance for all trials was evaluated using mean precision-recall (PR) curves. These curves were generated by interpolating precision values across 100 equally spaced recall points (0 to 1) and averaging across all 100 iterations for each minimum-data value. The resulting MDF threshold was then applied to the independent case study test datasets to evaluate its performance (Section 2.3).

2.2.2 *Configure species-specific filter*

To reduce reliance on overly restrictive classification score thresholds and time-intensive manual validation, we developed a species-specific filter (SSF) to exclude acoustic events inconsistent with false killer whale vocalizations prior to classification. The SSF criteria were empirically derived from the same visually verified false killer whale events described in Section 2.2.1 by establishing cut-off thresholds that encompassed the majority of observed data. Specifically, three criteria were defined: (1) an upper-bound threshold for median click peak frequency, (2) an upper-bound threshold for median whistle beginning frequency, and (3) a lower-bound threshold for the proportion of detected whistles exhibiting the species' stereotypic beginning frequency (4–8 kHz). Applying these filters increased the confidence that events entering classification met biologically informed false killer whale criteria, increasing confidence in lower classification scores while reducing the need for excessive manual review or reliance on conservative classification score thresholds.

Together, the MDF and SSF establish a sequential pre-classification framework. The first step utilizes the MDF to determine if an acoustic event possesses sufficient data to be reliably evaluated. For events meeting this minimum data requirement, the second step applies the SSF to ensure their acoustic characteristics align with those of false killer whales. Ultimately, if an

acoustic event fails either of these sequential filters, it is excluded from the dataset and is not included in the classification process.

2.3 Test pre-classification data filters

We assessed the performance of the pre-classification data filters (MDF and SSF) on acoustic events from towed array datasets that were systematically collected during cetacean line-transect abundance surveys conducted in 2017, 2023, and 2024 throughout the Hawaiian Islands (Table S1). The full case study dataset comprised visually verified events as well as acoustic-only events of odontocetes, which underwent manual review to verify the species identities assigned during real-time data collection. By applying the Pc-vs-Other model to both unfiltered and filtered events, we calculated multiple performance metrics to evaluate how the filters influenced classification results at different thresholds, including false positive and false negative rates, precision, and recall. In addition to comparing filtered and unfiltered results, we evaluated model performance using the full case study dataset with all events and a subset of the dataset restricted to the visually-verified events, which we designated as the ground truth dataset because their species identification was confirmed by visual observation. This comparison allowed us to directly quantify true positives and false negatives with and without the pre-classification filters.

3. Results

3.1 Data filters

3.1.1 Minimum-data filter

The sensitivity analysis used to establish the MDF included seven trials using unfiltered acoustic data, each using different combinations of signal types: 1) all signal types, 2) whistles, 3) clicks, 4) burst pulses, 5) clicks and whistles, 6) clicks and burst pulses, and 7) whistles and burst pulses. Results of trials 1-4 are included in Fig. 1 and trials 5-7 are provided in the supplementary material (Fig. S2).

Mean PR curves summarized classification performance for all trials across the 100 iterations for each minimum-data value. Overall, the trial with all signal types produced the highest and most consistent precision across all recall values compared to trials that only included a single signal type (Fig. 1). Within the all-signals trial, precision was notably lower across the recall range when using 3-10 signals per detector (Fig. 1a). Conversely, there were only modest gains in precision when using more than 20 signals per detector. In some instances, PR curves for 20 and 30,000 signals per detector converged, indicating that more data did not necessarily yield better classification results.

For trials that included only whistles or clicks (Fig. 1b, 1c), precision decreased gradually between recall values of 0.25 - 0.8, and precision for the clicks-only trial never exceeded 0.75 at higher recall values. However, including 20 signals per detector still achieved precision values comparable to those using larger amounts of data in both single-signal trials. The trial using only burst pulses performed poorly (Fig. 1d), with precision barely exceeding prevalence. Trials combining different signal types produced similar results to all-signal trials, with precision remaining stable at 0.78 - 0.81 across 20 signals per detector (Fig. S2). Based on these results, an MDF threshold of 20 signals per detector was established (when at least whistles and clicks were present).

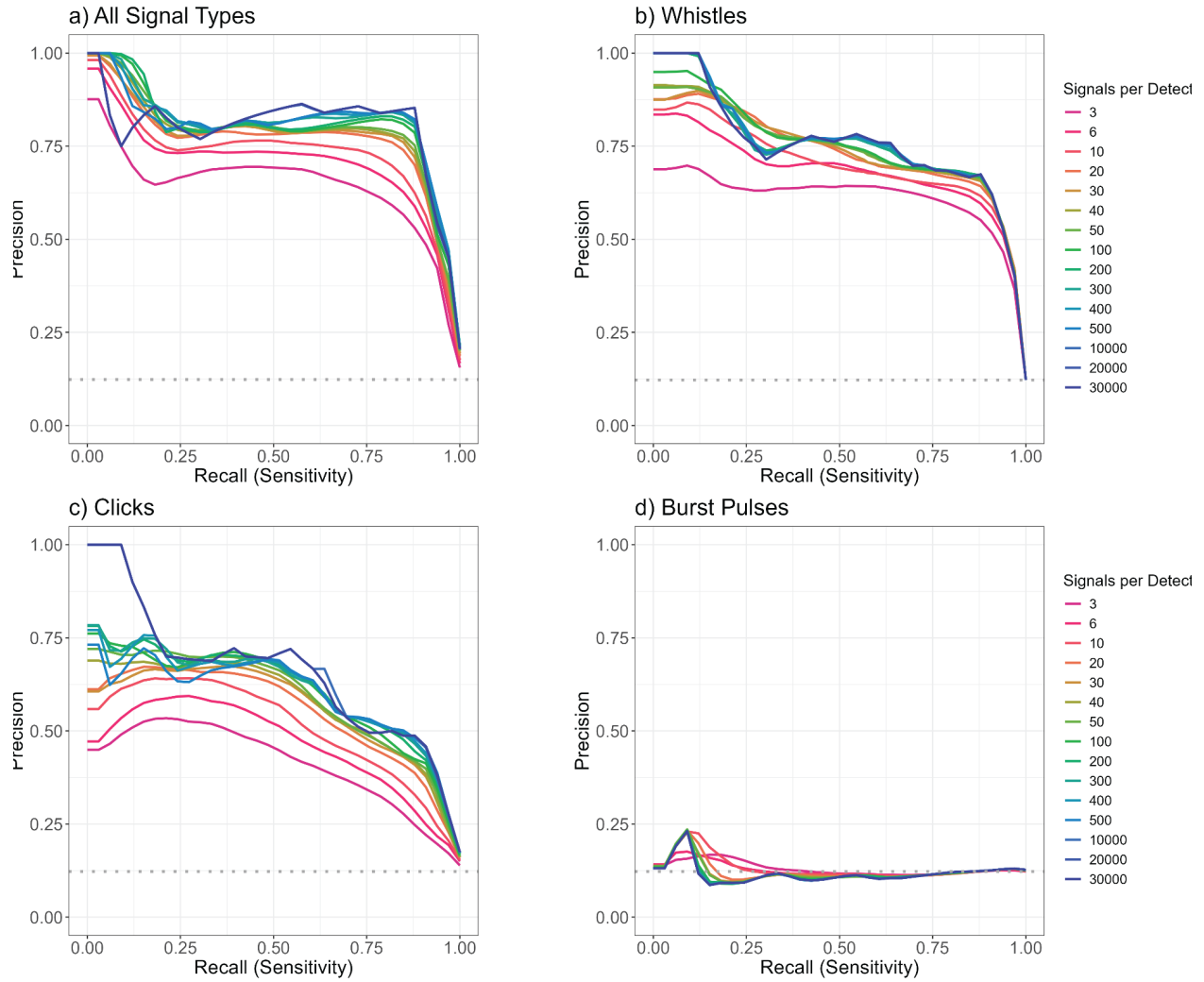


Fig. 1. Mean precision-recall curves showing the performance across 100 iterations for each minimum-data value (signals per detector) across four trials. Trials included datasets with a) all signal types, b) whistles, c) clicks, and d) burst pulses. For all trials, using 20 signals per detector achieved similar precision comparable to that obtained with larger amounts of data. Gray dotted lines represent prevalence, the proportion of the total acoustic events that belong to the positive class, false killer whale, in the dataset.

3.1.2 Species-specific filter

Analysis of the visually verified acoustic events yielded three final bounding criteria for the SSF to identify signals consistent with false killer whales (Fig. 2). Based on these data distributions, an acoustic event was required to exhibit a median click peak frequency of ≤ 22.4 kHz (the 95th percentile; Fig. 2a), a median whistle beginning frequency of ≤ 8.1 kHz (the 99th percentile; Fig. 2b), and at least a 0.50 proportion of whistles with a beginning frequency between 4 and 8 kHz (a lower bound encompassing 98% of the data; Fig. 1c). Events failing to meet any of these three criteria were not considered for classification.

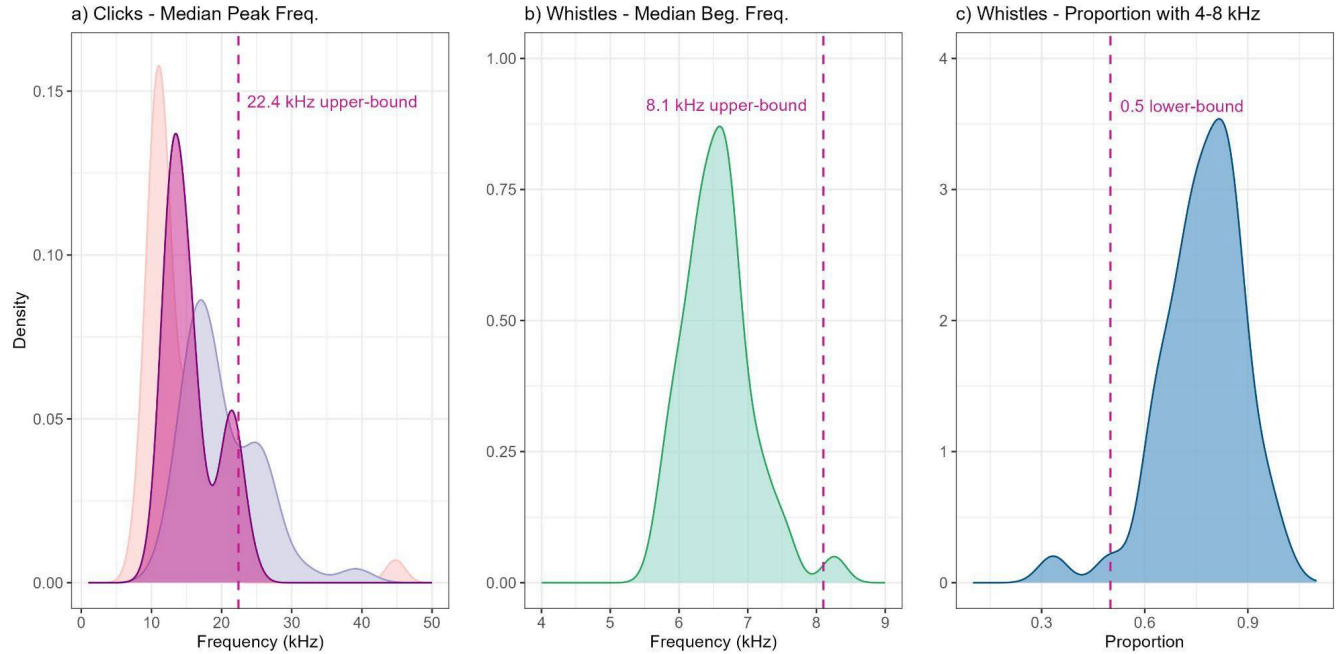


Figure 2. Distributions of acoustic characteristics derived from visually verified false killer whale acoustic events used to design the species-specific filter. Vertical dashed lines represent the cut-off thresholds for each criterion: (a) click median peak frequency (overall median peak frequency in dark pink, 25th percentile in light pink, and 75th percentile in light purple) with a 22.4 kHz upper bound representing the 95th percentile of the data; (b) median whistle beginning frequency with an 8.1 kHz upper bound representing the 99th percentile; and (c) proportion of

whistles with a beginning frequency between 4 and 8 kHz, with a 0.5 lower bound encompassing 98% of the data.

3.2 Evaluating Pre-classification Filter Performance

The three independent towed array case study datasets resulted in a combined total of 1,062 acoustic events with 24.2% visually verified to species and 75.8% acoustically detected only by the towed hydrophone array. Of the 258 visually-verified events, 31 were false killer whales. After filtering the acoustic data for general noise, 1,931,851 total signals were detected by the PAMGuard detectors, including 109,832 whistles, 1,780,071 clicks, and 41,896 burst pulses.

The total number of acoustic events classified as false killer whales decreased as both the classification threshold and the two filtering steps were applied (Fig 3). This relationship is demonstrated with a mirror barplot where the positive y-axis displays events predicted to be false killer whales and the negative y-axis represents the corresponding events removed as the MDF and SSF filters were applied, both individually and in combination (Fig 3). In the unfiltered dataset (with zero events removed prior to classification), the lowest classification score threshold of 0.5 yielded 133 predicted events. The most restrictive threshold of 0.9 reduced the number of predicted events to 12, pruning 91% of the events from the total dataset.

When the MDF and SSF filters were applied separately at the 0.5 threshold, the total predicted events were reduced to 85 and 80, respectively. As with the unfiltered dataset, increasing the threshold to 0.9 dropped the number of events considerably to 12 with only the MDF applied and 11 with the SSF. The most substantial reduction occurred when the MDF and

SSF filters were applied together, resulting in a total of 55 predicted events at the 0.5 threshold, representing a 59% reduction in data from the unfiltered dataset. The impact of the pre-classification data filters was most pronounced at lower classification thresholds, specifically between 0.5 and 0.7. Conversely, restricting the threshold to 0.8 or 0.9 led to a convergence where the number of predicted false killer whale events was similar, regardless of the filters applied (Fig. 3).

Table 1 provides quantitative metrics comparing classification performance between the unfiltered and filtered datasets. In the unfiltered visually verified dataset, applying the 0.5 threshold resulted in a false positive rate (FPR) of 4%, with nine other odontocete events misclassified as false killer whales. This FPR decreased to 0% only when the threshold was increased to 0.8 or higher. Additionally, one true false killer whale event was consistently misclassified across all thresholds due to recording noise (a false negative). When both pre-classification filters were applied, three visually verified false killer whale events were removed because their acoustic features fell outside the SSF boundaries. However, the false positives decreased from nine to zero, increasing the overall precision from 77% to 100%. While this marginally increased the false negative rate (FNR), it successfully eliminated all false positives, an expected trade-off inherent in using 98th and 99th percentile thresholds (Fig. 2). The FPR remained at 0% across more restrictive thresholds, but the FNR increased as the lower-scoring true positive events were excluded by the model.

The most substantial improvement occurred in the full dataset (with all events) at the 0.5 threshold, where false positives decreased from 75 to 3 and increased precision from 44% to 95%. While the SSF was designed to be as inclusive as possible, seven true-positive events (four visually verified and three acoustic-only) were excluded. These events represent biological

outliers that fell outside the established SSF criteria despite efforts to capture the full breadth of the species' acoustic repertoire. However, this marginal increase in the FNR is outweighed by the drastic reduction in false positives. Ultimately, applying the pre-classification filters to the full dataset at the 0.5 threshold yielded 21 confidently classified acoustic-only events as false killer whales representing a 68% increase in documented encounters from the 31 visually verified events alone.

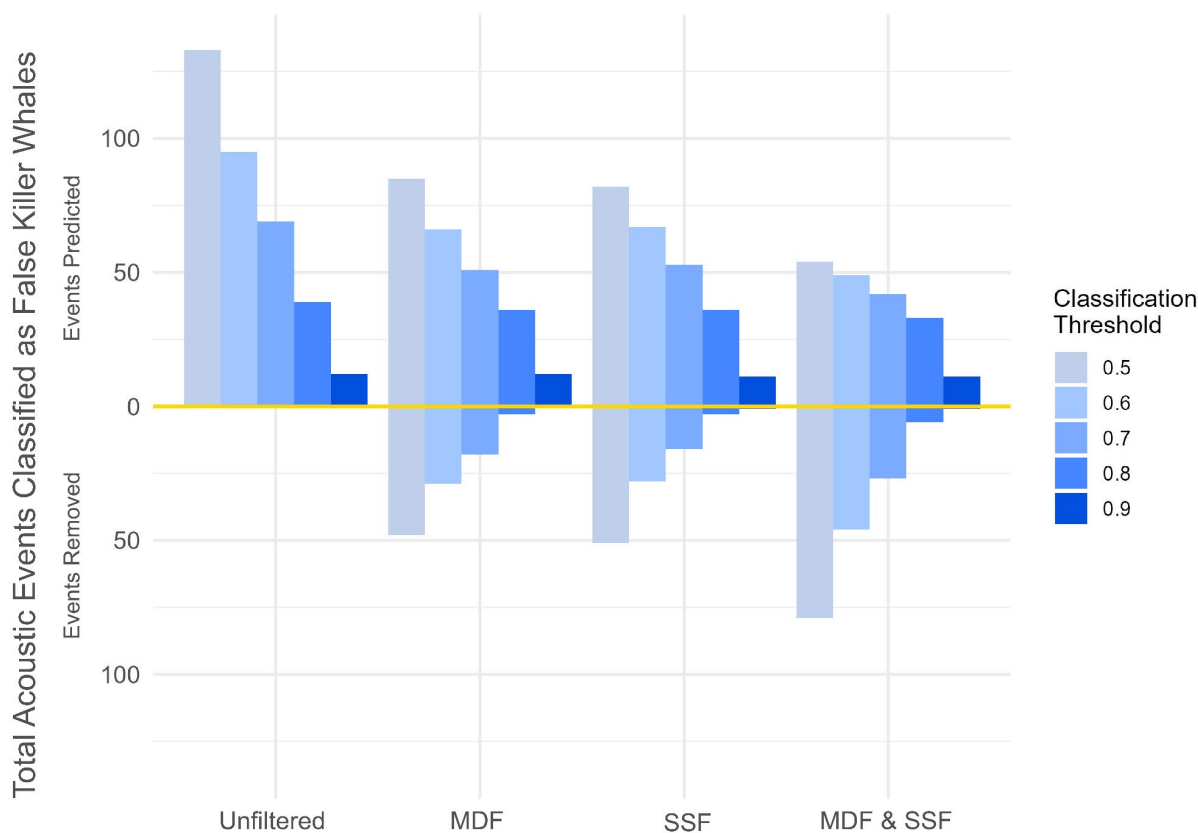


Figure 3. The number of acoustic events predicted as false killer whales from three towed array case study datasets (2017, 2023, 2024) are represented by the bars above the yellow line and

include visually-verified and acoustic-only events. The total events classified varied depending on the classification threshold and which pre-classification data filters were applied (MDF-minimum data filter, SSF - species-specific filter, and MDF & SSF combined). The baseline ‘Unfiltered’ group represents the total acoustic events classified as false killer whales with no data filters applied. As the classification threshold becomes more restrictive (lighter blue to darker blue represents lower to higher classification threshold, respectively) and one or both data filters are implemented, the number of acoustic events decreases and the number of events removed by the data filters (bars below the yellow line) increases, especially for thresholds of 0.5 - 0.7.

Table 1. Classification performance metrics for visually-verified events and all events combined between the unfiltered and filtered case study datasets. The filtered dataset applied both the minimum-data and species-specific filters. Predicted = number of events predicted as false killer whale, TP = true positive, FP = false positive, TN = true negative, FN = false negative, FPR = false positive rate, FNR = false negative rate.

Classification Threshold	Visually-Verified Events										All Events (Visually-Verified & Acoustic Only)								
	Dataset	Predicted	TP	FP	TN	FN	FPR	FNR	Precision	Recall	Predicted	TP	FP	TN	FN	FPR	FNR	Precision	Recall
0.5	Unfiltered	39	30	9	218	1	4%	3%	77%	97%	133	58	75	928	1	7%	2%	44%	98%
0.5	Filtered	27	27	0	40	4	0%	13%	100%	87%	55	52	3	116	7	3%	12%	95%	88%
0.6	Unfiltered	33	30	3	218	1	1.4%	3%	91%	97%	95	53	42	928	6	4%	10%	56%	90%
0.6	Filtered	27	27	0	40	4	0%	13%	100%	87%	48	47	1	116	12	1%	20%	98%	80%
0.7	Unfiltered	30	29	1	218	2	0.5%	6%	97%	94%	69	46	23	928	13	2%	22%	67%	78%
0.7	Filtered	26	26	0	40	5	0%	16%	100%	84%	40	40	0	116	19	0%	32%	100%	68%
0.8	Unfiltered	23	23	0	218	8	0%	26%	100%	74%	39	35	4	928	24	0%	41%	90%	59%
0.8	Filtered	22	22	0	40	9	0%	29%	100%	71%	31	31	0	116	28	0%	47%	100%	53%
0.9	Unfiltered	10	10	0	218	21	0%	68%	100%	32%	12	12	0	928	47	0%	80%	100%	20%
0.9	Filtered	10	10	0	40	21	0%	68%	100%	32%	11	11	0	116	48	0%	81%	100%	19%

4. Discussion

As management and conservation decisions increasingly rely on species classification from PAM data, extracting reliable, objective results from large datasets remains a bottleneck. For researchers monitoring rare species using automated classifiers, they must often choose between applying a high classification threshold (e.g., 0.9) to eliminate false positives at the cost of losing valuable true detections, or use a lower threshold (e.g., 0.5) to capture more events, which then requires a massive, subjective, manual review process that is difficult to reproduce. The pre-classification filtering steps developed in this study address this trade-off by objectively restricting candidate acoustic events based on data-driven criteria prior to classification. In our case study, this approach eliminated the need to manually review 75 non-target events at the 0.5 threshold. The overall classification precision increased while simultaneously adding 21 high-confidence, acoustic-only detections to the dataset, increasing the usable data by 68%. This structured workflow provides a reproducible approach that generates reliable results for downstream applications, such as species density and distribution modeling.

The success of this workflow relies on the sequential application of the MDF and SSF. The sensitivity analysis demonstrated that the MDF acts as a critical data-quality baseline by requiring a minimum of 20 signals per detector across all available signal types. This threshold consistently yielded precision values comparable to runs that included substantially larger volumes of data (Fig. 1). The noticeable drop in precision when fewer than 20 signals per detector were used, and when only certain signal types were included, highlights the importance of retaining sufficient signal counts to confidently distinguish a relatively rare event such as a false killer whale from other species.

Relying on the unique acoustic characteristics of visually verified encounters, the SSF

criteria proved consistent with existing published values for false killer whales (Oswald et al., 2007; Baumann-Pickering et al., 2015) and leveraged the distinctive features that drive the high accuracy of the Pc-vs-Other model (McCullough et al., 2021). While the SSF was designed to capture the vast majority of false killer whale encounters (Fig. 2), biological outliers will inevitably occur, as evidenced by the exclusion of four visually verified events and three acoustic-only events at the 0.5 threshold. Upon further review, these missed events contained noise from electrical interference or snapping shrimp which skewed the click peak frequency measurement, or the events narrowly exceeded the SSF boundaries due to the false killer whale signals themselves. Expanding the SSF boundaries to encompass these specific edge cases was not pursued, as doing so would inherently introduce more false positives from other highly vocal odontocete species, ultimately compromising the reliability of the filtered data. An additional workflow step could incorporate targeted manual review of such edge cases if deemed necessary by project-specific goals. However, in the context of population monitoring where high confidence in species classification is required, the significant improvement in precision provided by the filtering framework justifies this marginal loss in recall.

Developing this workflow highlighted the inherent challenges of automating biological data processing, as variability in acoustic signals will inevitably include outliers that fall outside standard statistical boundaries. Furthermore, unexpected noise can corrupt click detections, causing valid events to fail the filters even if the classification model still assigns them a high score. Additionally, this workflow does not currently account for mixed-species encounters. Because the underlying model was trained on single-species events, mixed groups would likely be excluded by the filters, potentially introducing bias that future studies will need to address. While these complexities mean we cannot completely eliminate manual analysis, this framework

provides a reproducible way to filter out the vast majority of non-target data.

This study demonstrates that coupling an event-level pre-classification filter with acoustic classification provides an objective, scalable method for analyzing large PAM datasets. A key strength of this framework is its flexibility; it can be readily adapted to other target species and classifiers, provided that robust, species-specific acoustic characteristics are defined and an accurate acoustic classification model is available. Future applications should evaluate the framework's robustness across broader geographic and temporal scales, and test its efficacy on datasets collected from other recording platforms, such as moored autonomous instruments or underwater gliders. While absolute certainty in acoustic classification may remain unattainable, this approach objectively refines the data to maximize classification precision, delivering the reliable information necessary to support effective conservation and management decisions.

5. Conclusion

This study introduces a standardized data-filtering framework for acoustic events designed to improve the accuracy of automated acoustic classification for North Pacific false killer whales. By implementing a minimum-data filter and a species-specific filter, researchers can significantly reduce false positive classifications in large passive acoustic datasets. This methodology allows for more reliable species identification when visual verification is not possible, effectively increasing the volume of usable data for population monitoring. The findings demonstrate that pre-classification filtering minimizes the need for time-consuming manual reviews while maintaining high precision. Ultimately, this reproducible workflow offers a scalable solution for cetacean passive acoustic data workflows that support conservation and management decisions.

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Supplementary Material

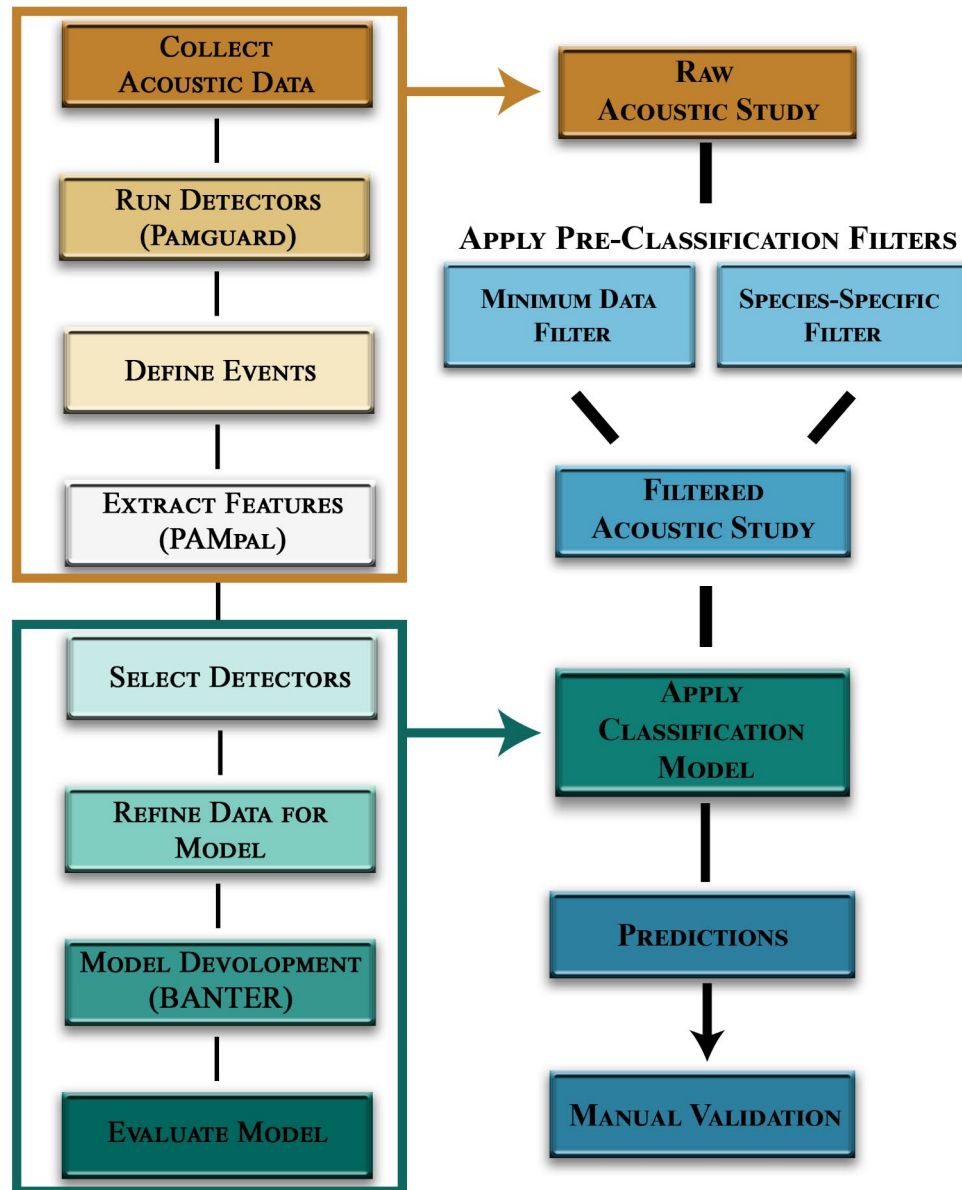
See supplementary material for additional figures and table providing more information for the methods and results.

373 Table S1. Technical specifications of the cetacean surveys conducted by the NOAA Pacific
374 Islands Fisheries Science Center included in this analysis.

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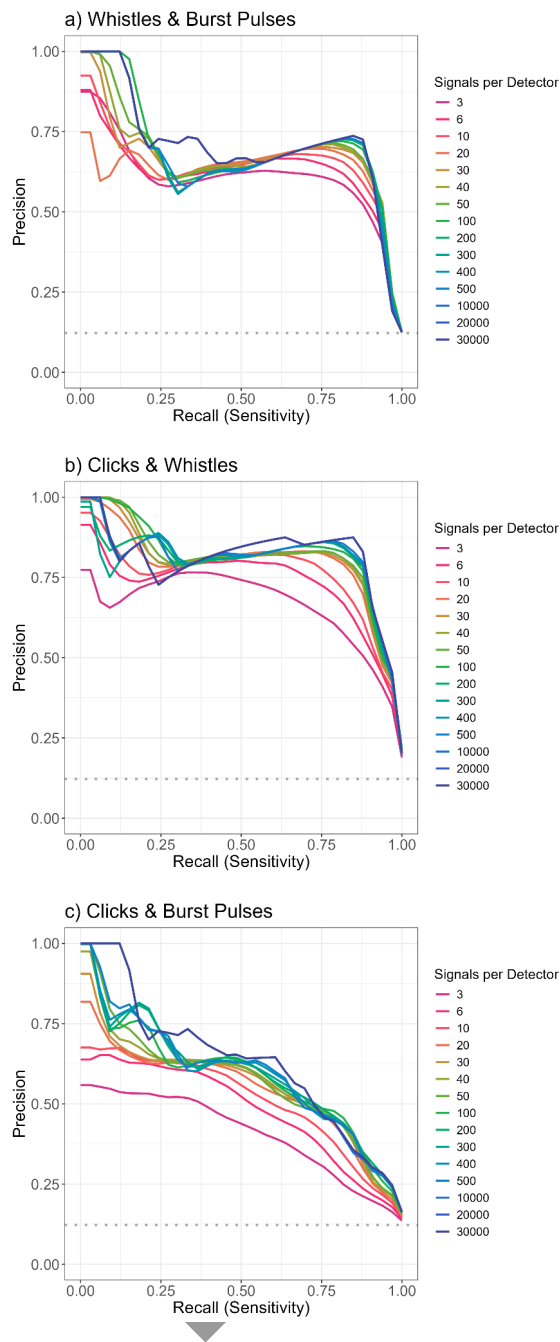
Survey Year	Study Region	Cruise Number(s)	Hydrophone Type	Hydrophone Frequency Response	Sampling Rate	Total Recording Hours	PAMGuard Version for Detection	Total Encounters for Sensitivity Analysis	Visually-Verified Acoustic Pseudorca Events	References for further survey information
2010	Hawaiian EEZ	1641; 1642	EDO EC65	2 - 40 kHz	192 kHz	1445	2.02.7b	8; 1	8; 1	Bradford, A. L., Olsson, E. M., Forney, K. A., Moore, J. E., & Barlow, J. (2021). Line-transect abundance estimates of cetaceans in US waters around the Hawaiian Islands in 2002, 2010, and 2017. NOAA Technical Memorandum NMFS-PFSC-115.
2011	Oahu to Palmyra	1108	EDO EC65	2 - 40 kHz	192 kHz	253	2.02.7b	2	2	NA
2012	American Samoa - Palmyra Atoll - Oahu	1203	EDO EC65	2 - 40 kHz	192 kHz	263	2.02.7b	1	1	Pacific Islands Fisheries Science Center (U.S.) Cruise report. (2012). NOAA Ship Oscar Elton Sate, Cruise SE-12-03, 23 April-17 May 2012, Palmyra Atoll Exclusive Economic Zone (EEZ), https://repository.library.noaa.gov/view/noaa/9070 .
2013	Papahānaumokuākea Marine National Monument	1303	APC 42-1021	2 - 40 kHz	192 kHz	293	2.02.7b	2	2	Bardey, Y., Olsson, E. M., Oswald, J. N., & Frankin, E. C. (2019). Whale classification of sympatric false killer whale populations in Hawaiian waters yields low accuracy rates. <i>Frontiers in Marine Science</i> , 6, 645.
2015	Marianas Archipelago	1502	HTL96-MIN	2 - 85 kHz	500 kHz	306	2.02.7b	2	2	Hill, M. C., Olsson, E. M., Bradford, A. L., Martien, K. K., Steel, D., & Balser, C. S. (2020). Assessing cetacean populations in the Mariana Archipelago: A summary of data and analyses arising from Pacific Islands Fisheries Science Center surveys from 2010–2019. NOAA Technical Memorandum, (NMFS-PFSC-108), 98. https://doi.org/10.25923/WRYE-6H14
2016	Main Hawaiian Islands	1604	HTL96-MIN	2 - 85 kHz	500 kHz	350	2.02.7b	1	1	Bardey, Y., Olsson, E. M., Oswald, J. N., & Frankin, E. C. (2019). Whale classification of sympatric false killer whale populations in Hawaiian waters yields low accuracy rates. <i>Frontiers in Marine Science</i> , 6, 645.
2017	Hawaiian EEZ	1705; 1706	HTL96-MIN	2 - 85 kHz	500 kHz	1743	2.00.16	NA	10; 11	Yano, K. M., Olsson, E. M., Keating, J. L., Ballance, L. T., Hill, M. C., Bradford, A. L., Allen, A. N., Joyce, T. W., Moom, J. E., & Henry, A. E. (2018). Cetacean and Seabird Data Collected during the Hawaiian Islands Cetacean and Ecosystem Assessment Survey (HICEAS), July–December 2017. NOAA Technical Memorandum NMFS-PFSC-72.
2018	Marianas Archipelago	1803	HTL96-MIN	2 - 85 kHz	500 kHz	339	2.01.05	27	0	Hill, M. C., Olsson, E. M., Bradford, A. L., Martien, K. K., Steel, D., & Balser, C. S. (2020). Assessing cetacean populations in the Mariana Archipelago: A summary of data and analyses arising from Pacific Islands Fisheries Science Center surveys from 2010–2019. NOAA Technical Memorandum, (NMFS-PFSC-108), 98. https://doi.org/10.25923/WRYE-6H14
2019	Main Hawaiian Islands / Papahānaumokuākea Marine National Monument	1901	HTL96-MIN	2 - 85 kHz	500 kHz	109	2.01.05	10	2	NA
2020	Main Hawaiian Islands	2001	HTL96-MIN	2 - 85 kHz	500 kHz	475	2.01.05	127	4	Yano, K. M., Olsson, E. M., McCullough, J. L., Hill, M. C., & Henry, A. E. (2020). Cetacean and seabird data collected during the Winter Hawaiian Islands Cetacean and Ecosystem Assessment Survey (Winter HICEAS), January–March 2020. NOAA Technical Memorandum NMFS-PFSC-111.
2021	Marianas Archipelago	2102	HTL96-MIN	2 - 85 kHz	500 kHz	339	2.01.05	89	10	Yano, K. M., Hill, M. C., Olsson, E. M., McCullough, J. L., & Henry, A. E. (2022). Cetacean and Seabird Data Collected During the Mariana Archipelago Cetacean Survey (MACS), May–July 2021. NOAA Technical Memorandum NMFS-PFSC-128.
2023	Hawaiian EEZ	2303; 2401	HTL96-MIN	2 - 85 kHz	500 kHz	1168	2.02.15	NA	5	Yano, K. M., Olsson, E. M., McCullough, J. L., Hill, M. C., Bradford, A. L., Joyce, T. W., ... & Henry, A. E. (2025). Cetacean and Seabird Data Collected During the Hawaiian Islands Cetacean and Ecosystem Assessment Survey (HICEAS), July–December 2023. NOAA Technical Memorandum NMFS-PFSC-175.
2024	Southeast of Hawaiian EEZ	2404	HTL96-MIN	2 - 85 kHz	500 kHz	389	2.02.15	NA	5	NA

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Fig. S1 The flow chart illustrates where the pre-classification data filters (light blue boxes) fit into the original workflow for acoustic data processing and classification model development described in Fig 1. of McCullough et al., 2021.



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Fig. S2 Mean precision-recall curves showing the performance across 100 iterations for each minimum-data value (signals per detector) across three trials with different combinations of signal types. Trials included datasets with a) whistles and burst pulses, b) clicks and whistles,

and c) clicks and burst pulses. Gray dotted lines represent prevalence, the proportion of the total acoustic events that belong to the positive class, false killer whale, in the dataset.

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Author Declarations

The authors declare that they have no conflicts of interest.

Data Availability

The data and code that support the findings of this study are openly available in a Github repository, BANTER_Pseudorca.

References

- Backus, R., & Schevill, W. (1966). Physeter clicks. *Whales, Dolphins and Porpoises*, 510–528.
- Baumann-Pickering, S., McDonald, M. A., Simonis, A. E., Solsona Berga, A., Merkens, K. P. B., Oleson, E. M., Roch, M. A., Wiggins, S. M., Rankin, S., Yack, T. M., & Hildebrand, J. A. (2013). Species-specific beaked whale echolocation signals. *The Journal of the Acoustical Society of America*, 134(3), 2293–2301. <https://doi.org/10.1121/1.4817832>
- Baumann-Pickering, S., Simonis, A. E., Oleson, E. M., Baird, R. W., Roch, M., & Wiggins, S. M. (2015). False killer whale and short-finned pilot whale acoustic identification. *Endangered Species Research*, 28(2), 97–108. <https://doi.org/10.3354/esr00685>
- Bittle, M., & Duncan, A. (2013). A review of current marine mammal detection and classification algorithms for use in automated passive acoustic monitoring. *Proceedings of Acoustics 2013 - Victor Harbor*, 1–8.
- Bradford, A. L., Forney, K. A., Oleson, E. M., & Barlow, J. (2014). Accounting for Subgroup Structure in Line-Transect Abundance Estimates of False Killer Whales (*Pseudorca crassidens*) in Hawaiian Waters. *PLoS ONE*, 9(2), e90464. <https://doi.org/10.1371/journal.pone.0090464>
- Davis, G. E., Baumgartner, M. F., Bonnell, J. M., Bell, J., Berchok, C., Bort Thornton, J., Brault, S., Buchanan, G., Charif, R. A., Cholewiak, D., Clark, C. W., Corkeron, P., Delarue, J., Dudzinski, K., Hatch, L., Hildebrand, J., Hodge, L., Klinck, H., Kraus, S., ... Van Parijs, S. M. (2017). Long-term passive acoustic recordings track the changing distribution of North Atlantic right whales (*Eubalaena glacialis*) from 2004 to 2014. *Scientific Reports*, 7(1), 13460. <https://doi.org/10.1038/s41598-017-13359-3>
- Gillespie, D., Macaulay, J., Oswald, M., & Roch, M. (2026). PAMGuard: Application software for passive acoustic detection, classification, and localisation of animal sounds. *The Journal of the Acoustical Society of America*, 159(1), 437–443. <https://doi.org/10.1121/10.0042245>
- Gruden, P., Barkley, Y. M., & McCullough, J. L. K. (2023). Vocal behavior of false killer whale (*Pseudorca crassidens*) acoustic subgroups. *Frontiers in Marine Science*, 10, 1147670. <https://doi.org/10.3389/fmars.2023.1147670>

Johnson, M., Aguilar De Soto, N., & Madsen, P. (2009). Studying the behaviour and sensory ecology of marine mammals using acoustic recording tags: A review. *Marine Ecology Progress Series*, 395, 55–73. <https://doi.org/10.3354/meps08255>

Kowarski, K. A., & Moors-Murphy, H. (2021). A review of big data analysis methods for baleen whale passive acoustic monitoring. *Marine Mammal Science*, 37(2), 652–673. <https://doi.org/10.1111/mms.12758>

Leroy, E. C., Thomisch, K., Royer, J.-Y., Boebel, O., & Van Opzeeland, I. (2018). On the reliability of acoustic annotations and automatic detections of Antarctic blue whale calls under different acoustic conditions. *The Journal of the Acoustical Society of America*, 144(2), 740–754. <https://doi.org/10.1121/1.5049803>

Marques, T. A., Thomas, L., Martin, S. W., Mellinger, D. K., Ward, J. A., Moretti, D. J., Harris, D., & Tyack, P. L. (2013). Estimating animal population density using passive acoustics. *Biological Reviews*, 88(2), 287–309. <https://doi.org/10.1111/brv.12001>

McCullough, J. L. K., Wren, J. L. K., Oleson, E. M., Allen, A. N., Siders, Z. A., & Norris, E. S. (2021). An Acoustic Survey of Beaked Whales and *Kogia* spp. In the Mariana Archipelago Using Drifting Recorders. *Frontiers in Marine Science*, 8, 664292. <https://doi.org/10.3389/fmars.2021.664292>

Miller, B. S., Madhusudhana, S., Aulich, M. G., & Kelly, N. (2023). Deep learning algorithm outperforms experienced human observer at detection of blue whale D-calls: A double-observer analysis. *Remote Sensing in Ecology and Conservation*, 9(1), 104–116.

Oswald, J. N., Rankin, S., Barlow, J., & Lammers, M. O. . (2007). A tool for real-time acoustic species identification of delphinid whistles. *The Journal of the Acoustical Society of America*, 122(1), 587–595. <https://doi.org/10.1121/1.2743157>

Rankin, S., Archer, F., Keating, J. L., Oswald, J. N., Oswald, M., Curtis, A., & Barlow, J. (2017). Acoustic classification of dolphins in the California Current using whistles, echolocation clicks, and burst pulses. *Marine Mammal Science*, 33(2), 520–540. <https://doi.org/10.1111/mms.12381>

Rankin, S., Oswald, J. N., & Barlow, J. (2008). Acoustic behavior of dolphins in the Pacific Ocean: Implications for using passive acoustic methods for population studies. *Canadian Acoustics*, 36(1), 88–92.

Rice, A., Širović, A., Trickey, J. S., Debach, A. J., Gottlieb, R. S., Wiggins, S. M., Hildebrand, J. A., & Baumann-Pickering, S. (2021). Cetacean occurrence in the Gulf of Alaska from long-term passive acoustic monitoring. *Marine Biology*, 168(5), 72. <https://doi.org/10.1007/s00227-021-03884-1>

Ross, S. R. P. -J., O'Connell, D. P., Deichmann, J. L., Desjonquères, C., Gasc, A., Phillips, J. N., Sethi, S. S., Wood, C. M., & Burivalova, Z. (2023). Passive acoustic monitoring provides a fresh perspective on fundamental ecological questions. *Functional Ecology*, 37(4), 959–975. <https://doi.org/10.1111/1365-2435.14275>

Sakai, T. (2020). *PAMpal* (Version 1.2.1) [R]. <https://doi.org/10.5281/zenodo.12728738>

Sayigh, L., Daher, M. A., Allen, J., Gordon, H., Joyce, K., Stuhlmann, C., & Tyack, P. (2016). The Watkins Marine Mammal Sound Database: An online, freely accessible resource. *Proceedings of Meetings on Acoustics*, 27(1), 040013. <https://doi.org/10.1121/2.0000358>

Širović, A. (2016). Variability in the performance of the spectrogram correlation detector for North-east Pacific blue whale calls. *Bioacoustics*, 25(2), 145–160. <https://doi.org/10.1080/09524622.2015.1124248>

Usman, A. M., Ogundile, O. O., & Versfeld, D. J. J. (2020). Review of Automatic Detection and Classification Techniques for Cetacean Vocalization. *IEEE Access*, 8, 105181–105206. <https://doi.org/10.1109/ACCESS.2020.3000477>

Van Parijs, S., Clark, C., Sousa-Lima, R., Parks, S., Rankin, S., Risch, D., & Van Opzeeland, I. (2009). Management and research applications of real-time and archival passive acoustic sensors over varying temporal and spatial scales. *Marine Ecology Progress Series*, 395, 21–36. <https://doi.org/10.3354/meps08123>

Ziegenhorn, M. A., Frasier, K. E., Hildebrand, J. A., Oleson, E. M., Baird, R. W., Wiggins, S. M., & Baumann-Pickering, S. (2022). Discriminating and classifying odontocete echolocation clicks in the Hawaiian Islands using machine learning methods. *PLOS ONE*, 17(4), e0266424. <https://doi.org/10.1371/journal.pone.0266424>

Zimmer, W. M. X., Tyack, P. L., Johnson, M. P., & Madsen, P. T. (2005). Three-dimensional beam pattern of regular sperm whale clicks confirms bent-horn hypothesis. *The Journal of the Acoustical Society of America*, 117(3), 1473–1485. <https://doi.org/10.1121/1.1828501>